

Differences in striatal dopamine transporter density between tremor dominant and non-tremor Parkinson's disease

Valtteri Kaasinen · Maija Kinos · Juho Joutsa ·
Marko Seppänen · Tommi Noponen

Received: 24 February 2014 / Accepted: 2 May 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract

Purpose Parkinson's disease (PD) can manifest with a tremor-dominant or a non-tremor (akinetic-rigid) phenotype. Although the tremor-dominant subtype may show a better prognosis, there is limited information on the phenotypic differences regarding the level of striatal dopamine transmission. The present study investigated striatal dopamine transporter (DAT) binding characteristics in a large sample of patients with and without tremor.

Methods [123 I]FP-CIT SPECT scans of 231 patients with a clinical diagnosis of PD and abnormal FP-CIT binding (157 with tremor, 74 without tremor) and 230 control patients with normal FP-CIT binding (148 with tremor, 82 without tremor) were analysed using an automated region-of-interest analysis of the scans (BRASS). Specific striatal binding ratios were compared between phenotypes and groups using age, sex, and symptom duration, predominant side of symptoms, dopaminergic medications and scanner as covariates.

Results Patients with PD had 28.1 – 65.0 % lower binding in all striatal regions compared to controls ($p < 0.001$). The mean FP-CIT caudate nucleus uptake and the left caudate nucleus uptake were higher in PD patients with tremor than in PD patients without tremor (mean 9.0 % higher, left 10.5 % higher; $p < 0.05$), whereas there were no differences between tremor and non-tremor control patients. No significant effects

of tremor on DAT binding were observed in the anterior or posterior putamen.

Conclusion The motor phenotype is associated with the extent of caudate dopamine terminal loss in PD, as dopamine function is relatively more preserved in tremor patients. Symptom type is related to caudate dopamine function only in association with Parkinsonian dopaminergic degeneration, not in intact dopamine systems in patients with non-PD tremor.

Keywords Dopamine transporter · Parkinson's disease · Tremor · SPECT

Introduction

Tremor is a symptom characterized by involuntary, rhythmic and sinusoidal alternating movements of one or more body parts [1]. It is one of the cardinal symptoms of Parkinson's disease (PD) but seems to differ from bradykinesia and rigidity both clinically and pathophysiologically [2, 3]. Several lines of evidence indicate that tremor is a marker of relatively benign PD [3]. Motor progression [4–6] and cognitive progression [7] of tremor-dominant PD appear to be slower than progression of non-tremor PD, i.e. postural instability gait disorder (PIGD) or akinetic-rigid PD. The slower rate of progression in tremor-dominant PD is correlated with less severe nigral cell loss based on post-mortem examination [8]. Non-tremor PD patients show reduced brain activity in the prefrontal cortex and globus pallidus [9], more prominent grey matter atrophy in motor-related regions and decreased functional connectivity [10] compared to tremor-dominant patients. From a treatment-related perspective, non-tremor PD seems to be associated with an increased risk of levodopa-induced dyskinesias [11].

Although lower CSF concentrations of β -amyloid 1–42, T-tau, P-tau₁₈₁ and α -synuclein have been reported in patients

V. Kaasinen (✉) · M. Kinos · J. Joutsa
Division of Clinical Neurosciences, University of Turku and Turku
University Hospital, POB 52, 20521 Turku, Finland
e-mail: valtteri.kaasinen@tyks.fi

V. Kaasinen · M. Kinos · J. Joutsa · M. Seppänen
Turku PET Centre, University of Turku and Turku University
Hospital, Turku, Finland

M. Seppänen · T. Noponen
Department of Clinical Physiology and Nuclear Medicine,
University of Turku and Turku University Hospital, Turku, Finland

with the non-tremor PD subtype compared to tremor-dominant patients [12], data on possible imaging biomarkers in the nigrostriatal dopaminergic system are inconclusive. Previous studies with [^{123}I]FP-CIT ([^{123}I]N- ω -fluoropropyl-2 β -carbomethoxy-3 β -(4-iodophenyl)nortropine) SPECT have indicated that tremor-dominant patients may have higher striatal dopamine transporter (DAT) binding [13–16], although there are also opposite [17] and negative [18] results. The sample sizes in these studies have varied between 20 and 62 patients, of whom 10 to 24 patients have been tremor-dominant. In the largest study to date, Eggers et al. studied 122 PD patients (23 with tremor) and found no significant differences between subtypes when the effects of age, disease duration, disease severity and medication were excluded, although visual image analysis showed differences between groups [19].

It therefore appears that while there are clinical differences between tremor-dominant and non-tremor PD patients, the results of functional imaging studies of the DAT system are mixed but suggest a greater degree of neurodegeneration in non-tremor PD. It is possible that the discrepant results reflect methodological differences together with small sample sizes.

In the present study, we investigated differences in DAT density between tremor and non-tremor PD patients in the largest reported sample of patients, and conducted an automated analysis that was free from operator-induced bias. To investigate the role of neurodegeneration of the brain dopamine system in PD, we also included tremor and non-tremor control patients who had normal DAT scans and no evidence of PD.

Materials and methods

[^{123}I]FP-CIT SPECT scanning was performed at the Department of Nuclear Medicine, Turku University Hospital, Turku, Finland, over a 6-year period (2007 – 2012). We identified 231 patients with abnormal [^{123}I]FP-CIT scans in combination with a clinical diagnosis of idiopathic PD, and 230 patients with normal scans in combination with no evidence of PD or other neurodegenerative parkinsonism syndrome. The classification (abnormal/normal) was based on the initial clinical (visual with or without semiquantitative) judgement of a nuclear medicine physician. The study flow chart is presented in Fig. 1. The study protocol was approved by the ethics committee of the local hospital district and was conducted according to the principles of the Declaration of Helsinki.

A GE Infinia II Hawkeye SPECT/CT scanner with 3/8-inch thick crystals (GE Medical Systems, Milwaukee, WI) and a Picker Irix gamma camera with 3/4-inch thick crystals (Picker International, Uniontown, OH) were used for imaging. The thyroid was blocked 60 min before injection of tracer with an oral dose of 250 mg of KClO_4 (1 % solution). The injected

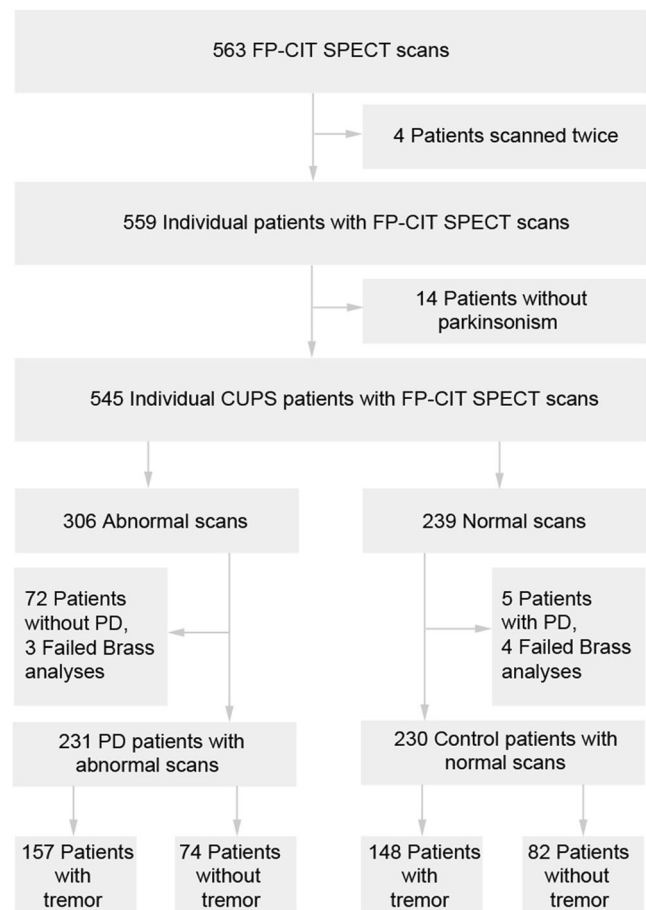


Fig. 1 Study flow chart. Of the 545 patients with clinically uncertain parkinsonian syndrome (CUPS), 231 with an abnormal [^{123}I]FP-CIT SPECT scan were diagnosed with PD and 230 with a normal [^{123}I]FP-CIT SPECT scan showed no evidence of PD or other neurodegenerative parkinsonism syndrome

activity of [^{123}I]FP-CIT was 185 MBq, and scanning was performed 4 h after injection. The data were acquired using LEHR collimators. A circular orbit with 120 projections and 25 s scanning time per view was acquired during 180° of rotation. A 64 × 64 matrix size, a zoom of 2.0 and a photopeak energy window at 159 keV ± 10 % were used. The SPECT images were reconstructed using the ordered subsets expectation algorithm of Hybrid Recon Neurology software (version 1.0.15) including Chang's attenuation ($\mu=0.146$ 1/cm), collimator and Monte-Carlo-based scatter correction with 16 iterations and four subsets (Hermes Medical Solutions AB, Stockholm, Sweden). A 3-D Gaussian postfilter with 0.7 cm full-width at half-maximum was used.

All scans were analysed using BRASS automated analysis software (version 3.6; Hermes Medical Solutions) by one of the investigators (M.K.). Coregistration of automatically drawn regions of interest and striatal structures were visually verified in each patient. Scanner-specific corrections were used for the BRASS analyses. Specific binding ratios (SBR) for six regions were calculated (right and left caudate, right

and left anterior putamen, and right and left posterior putamen) using the occipital cortex as the reference region: $SBR = (VOI_{\text{caudate or putamen}} - VOI_{\text{occipital}}) / VOI_{\text{occipital}}$ [20].

All data were collected by one of the investigators (V.K.) using individual clinical patient histories, hospital records and referrals together with the reanalysed SPECT images. The following factors were categorized: possible presence of Parkinsonism at the time of imaging, presence of any tremor type at the time of imaging, age and sex of the patient, duration of motor symptoms before scanning, possible dopaminergic medication at the time of imaging, predominant side of motor symptoms and the scanner. The clinical neurological diagnoses of the patients were collected in March 2013 (0.25–6.25 years after SPECT imaging) using the most recent patient files from the treating neurologist.

The differences between the tremor and non-tremor patients were compared between PD patients and controls (two groups, four subgroups) using a two-way analysis of variance (ANOVA) with age, sex, symptom duration, symptom side, scanner and medication as covariates. The primary analysis was performed with mean regional values [(left+right)/2], and individual hemispheric values (left, right, ipsilateral or contralateral) were compared if a significant difference was observed in the primary analysis. The level of statistical significance was set at $p < 0.05$.

Results

PD patients with tremor had a 9.0 % higher mean caudate tracer uptake than PD patients without tremor, whereas there were no differences between the tremor and non-tremor control patients (interaction effect group $\times$ subgroup $F=4.23$, $df=1$, $p=0.04$; Table 1, Fig. 2). This effect was also observed in the left caudate (10.5 % difference in the PD group, 0 % difference in the control group; interaction $F=4.59$, $df=1$, $p=0.03$) but not in the right caudate. No significant interactions were detected in the anterior putamen ($F=1.76$, $df=1$, $p=0.19$) or the posterior putamen ($F=0.03$, $df=1$, $p=0.85$). In the patients with lateralized symptoms (202 PD patients, 164 controls; patients with symmetrical symptoms excluded), no significant differences between tremor and non-tremor patients were seen in the ipsilateral or contralateral caudate ($p > 0.17$). The demographic characteristics of the patients together with the main ANOVA results are presented in Table 1. The difference in caudate tracer uptake between tremor dominant and non-tremor PD patients was also significant without any covariates (independent samples t test $p=0.004$).

In patients with PD, FP-CIT uptake was 28.1–34.3 % lower in the caudate ($F=359.5$, $df=1$, $p < 0.001$), 43.8–49.7 % lower in the anterior putamen ($F=792.8$, $df=$

1, $p < 0.001$) and 61.0–65.0 % lower in the posterior putamen ($F=1323.4$, $df=1$, $p < 0.001$) compared to uptake in controls.

Discussion

The results of the present large retrospective study indicate that the extent of caudate nucleus dopamine terminal loss is greater in non-tremor PD patients than in tremor PD patients, as investigated using DAT imaging approximately 2 years after symptom onset. The association was selectively related to Parkinsonian tremor and not to tremor in patients with normal striatal DAT binding. No significant differences were observed in the posterior putamen, which is the striatal region that is most severely affected by PD.

Loss of nigrostriatal dopamine neurons is the hallmark of PD, and the extent of degeneration correlates well with the severity of bradykinesia, and posture, gait and other midline symptoms [21]. However, one of the paradoxical aspects of Parkinsonian tremor is the weak relationship with dopamine deficiency [22]. The pathophysiology of Parkinsonian tremor is poorly understood, but there are data to support the theory that it is mediated primarily by cerebellothalamocortical pathways [14, 23], although basal ganglia networks are also involved [23, 24]. It is possible that a combination lesion in the nigrostriatal and cerebellothalamocortical circuits is required for PD tremor to emerge, and nigral loss alone is insufficient to produce resting tremor [22]. Thus, the lesser degree of dopamine neuron loss in tremor patients in the present study could be viewed as a reflection of dysfunctional circuitry that has its major lesion outside the striatum, such as in the thalamus. Caudate dopamine function may not primarily be reflected in the motor symptoms, but rather in the non-motor differences between the two motor phenotypes. PD patients with the non-tremor phenotype seem to be at higher risk of cognitive impairment compared to tremor-dominant PD patients [7, 25], and they seem to have a greatly increased risk of dementia [26] compared to tremor-dominant patients. Non-tremor patients also tend to be more depressed than patients with the tremor-dominant phenotype [27]. Indeed, impairments in cognitive tests have been reported to correlate with losses in caudate dopamine function in PD [28], and an inverse correlation between depression scores and caudate DAT availability in PD has been reported [29]. Future SPECT studies that combine neuropsychological and neuropsychiatric measures should be conducted to investigate this theory in detail.

A possible explanation for the lesser degree of dopamine terminal cell loss in tremor patients is that they or their next of kin may become aware of motor symptoms earlier than patients with non-tremor early PD. Therefore, although the duration of manifested symptoms may be the same, the duration of premanifest disease could be longer in non-tremor patients. This difference in the underlying level of progression

Table 1 Main demographic characteristics of the studied patients with regional covariate-adjusted specific binding ratios of FP-CIT uptake and 95 % confidence intervals

	PD patients (abnormal FP-CIT uptake)		Controls (normal FP-CIT uptake)		<i>p</i> value ^a
	Tremor	No tremor	Tremor	No tremor	
No. of patients	157	74	148	82	–
Age (years), mean (SD)	67.0 (10.5)	66.0 (10.9)	65.6 (11.2)	61.8 (13.1)	–
Sex, <i>n</i>					
Male	83	49	70	42	–
Female	74	25	78	40	–
Symptom duration (years), mean (SD)	2.2 (2.8)	1.9 (2.1)	5.4 (8.6)	2.7 (3.1)	–
Symptom side, <i>n</i>					
Right	80	24	73	23	–
Left	70	28	48	20	–
Symmetrical	7	22	27	39	–
Medication, <i>n</i>					
Dopaminergic antiparkinsonian	31	15	19	8	–
Anti-dopaminergic (dopamine receptor antagonists)	5	3	33	12	–
None	121	56	96	62	–
Scanner, <i>n</i>					
I	67	27	50	28	–
II	58	27	63	29	–
III	32	20	35	25	–
Specific binding ratio, mean (range)					
Caudate	1.94 (1.86 – 2.03)	1.78 (1.67 – 1.89)	2.70 (2.62 – 2.78)	2.71 (2.61 – 2.81)	0.04*
Anterior putamen	1.65 (1.55 – 1.74)	1.50 (1.38 – 1.61)	2.98 (2.89 – 3.06)	2.94 (2.83 – 3.06)	0.19
Posterior putamen	0.89 (0.81 – 0.96)	0.82 (0.72 – 0.91)	2.34 (2.27 – 2.41)	2.28 (2.19 – 2.37)	0.85
Right caudate	1.88 (1.79 – 1.98)	1.75 (1.64 – 1.87)	2.63 (2.55 – 2.71)	2.65 (2.54 – 2.76)	0.10
Left caudate	2.00 (1.91 – 2.09)	1.81 (1.69 – 1.92)	2.77 (2.69 – 2.85)	2.77 (2.66 – 2.88)	0.03*

**p*<0.05

Putaminal values in the left, right, ipsilateral and contralateral hemispheres showed no significant differences

^a Two-way ANOVA interaction-effect with age, sex, symptom duration, symptom side, medication and scanner as covariates

could explain why non-tremor patients show lower DAT density in the early stages of the disease with seemingly similar symptom durations. However, the anatomical distribution of the DAT binding difference between tremor and non-tremor patients does not support this theory. One could expect to see the largest difference in the posterior putamen, not in the caudate nucleus, if non-tremor patients are more advanced at the time of SPECT imaging. Furthermore, there do not seem to be differences in the ages of onset between tremor-dominant and akinetic-rigid patients [30], which do not support a considerably longer period of unrecognized disease in non-tremor patients. Eggers et al. [19] investigated visual interpretation of the scans together with BRASS. They found no significant differences between tremor-dominant and non-tremor patients using BRASS after covariate correction. Instead they found that tremor patients (*n*=23) showed a more “eagle-wing shaped” striatal configuration (more normal, with

some putaminal binding) than non-tremor patients who showed a more “egg-shaped” striatal configuration (more abnormal, without putaminal binding). It is possible that not only the average regional tracer binding but the diagnostically typical appearance of the striatum or the shape of the hot spot is also important. Further comparative studies (visual vs. semiquantitative) are needed.

Several subtypes of PD have been suggested to exist, but one of the most commonly used phenotypic categorization of PD is tremor-dominant vs. non-tremor-dominant disease (either PIGD or akinetic-rigid) [6, 31]. Tremor-dominant, non-tremor-dominant and intermediate subtypes are most commonly defined as the sum of Unified Parkinson’s Disease Rating Scale (UPDRS) tremor items divided by the sum of UPDRS PIGD items [5, 32]. Patients have also been assigned to groups based on UPDRS item score distributions or expert judgement [31]. Due to the retrospective nature of the present

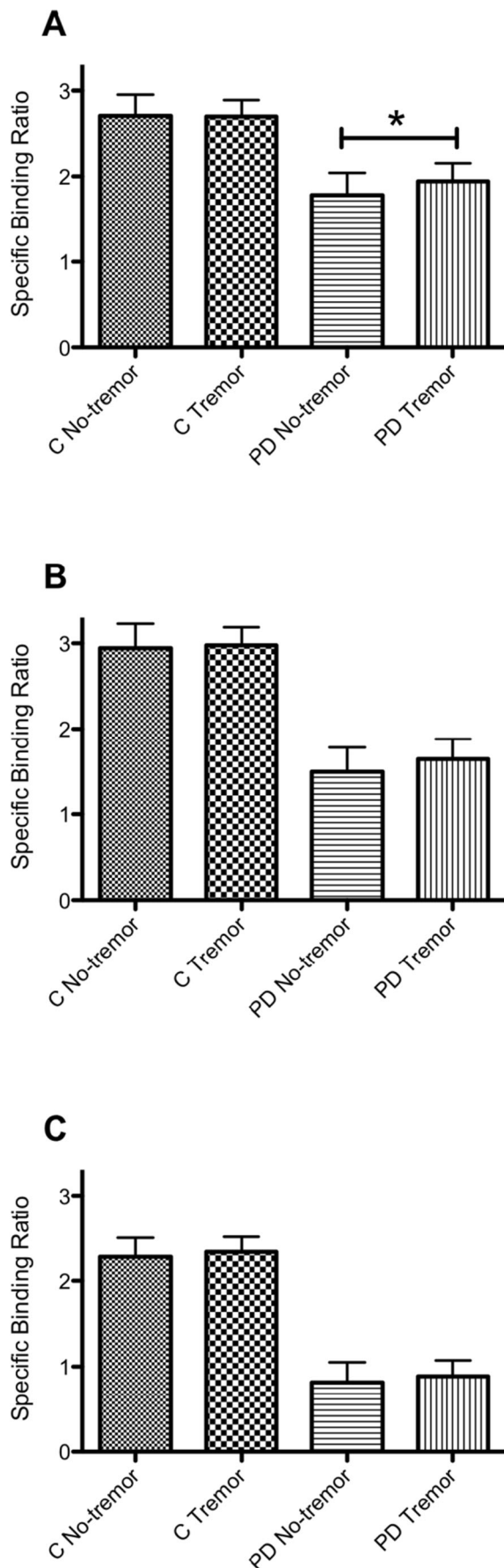


Fig. 2 Effect of tremor on FP-CIT binding in patients with PD compared to patients with normal FP-CIT binding (*C* controls). Estimated covariate-adjusted marginal means from the ANOVA model are presented with 95 % confidence intervals. **a** Mean (right + left) caudate nucleus binding (significant interaction effect, $p=0.04$). **b** Mean anterior putamen binding (no significant interaction effect). **c** Mean posterior putamen binding (no significant interaction effect)

study, UPDRS scores were not available for phenotypic categorization, and the assignment of the patients was based on the presence of any tremor at the time of imaging. Therefore, the tremor group in the present study included intermediate patients who fall between tremor-dominant and PIGD patients. The reported difference in the caudate DAT density could thus be an underestimation of the true effect if the differences between tremor-dominant and PIGD patients are considered. However, the present results are an accurate estimation of the differences between PD patients who have any tremor vs. PD patients who have no tremor.

One limiting factor in the interpretation of the present and all previous SPECT studies is the diagnostic accuracy of early parkinsonism syndromes. Although all clinical PD diagnoses were made by neurologists and all patients had pathological FP-CIT scans, there were most likely some incorrect clinical early PD diagnoses in our sample. The most likely diagnostic errors include patients with progressive supranuclear palsy (PSP) and parkinsonism variant of multiple system atrophy (MSA-P). Because the Richardson syndrome variant of PSP is not often associated with tremor [33], patients with misdiagnosed early Richardson syndrome are more likely to have been included in the non-tremor PD group than in the PD group. It has also been reported that PSP may show relatively more severe caudate DAT binding loss compared to PD [34]. If these two possibilities are considered together, one could speculate that the caudate DAT difference is partially caused by the inclusion of patients with Richardson syndrome who were misdiagnosed as non-tremor PD. However, the incidence rate of all tauopathies among patients with parkinsonism is clearly lower than synucleinopathies (1.1 vs. 21.0 per 100 000 person years) [35] and the effect of PSP in the present population may be marginal. Follow-up of the present patients for several years will determine how much of the effect is caused by misdiagnosed parkinsonism, and patients with changed diagnoses can be excluded. Tremor is not rare in MSA [36], and patients with misdiagnosed MSA are likely to have been more evenly distributed between the tremor and non-tremor groups. Ultimately, neuropathological confirmation of PD is needed.

Conclusion

Previous [123 I]FP-CIT SPECT studies comparing smaller samples of tremor-dominant and non-tremor patients have provided contradictory results [13–19]. The present results

indicate that PD patients with tremor have a less severe dopaminergic defect in the caudate nucleus compared to PD patients without tremor. This finding is consistent with the clinically more favourable prognosis of tremor PD. The localization of the findings to the caudate nucleus rather than to the posterior putamen suggests that the effect is not related to a longer unrecognized disease progression in non-tremor PD or to other major motor symptoms of the disease. The results contribute significantly to the understanding of the neurobiological basis of phenotypic differences in patients with PD. Possible therapeutic implications in different PD phenotypes, such as the risk of dyskinesias, remain unknown and could be a focus of further studies.

Ethical standards

The study protocol was approved by the ethics committee of the local hospital district and was conducted according to the principles of the Declaration of Helsinki.

Acknowledgments The contribution of the staff of the Department of Nuclear Medicine, Turku University Hospital, is gratefully acknowledged. This study was financially supported by the Academy of Finland (decision # 256836) and Turku University Hospital (ERVA funds).

Conflicts of interest None.

References

1. Abdo WF, van de Warrenburg BP, Burn DJ, Quinn NP, Bloem BR. The clinical approach to movement disorders. *Nat Rev Neurol*. 2010;6:29–37. doi:10.1038/nrneurol.2009.196.
2. Hallett M. Parkinson's disease tremor: pathophysiology. *Parkinsonism Relat Disord*. 2012;18 Suppl 1:S85–6. doi:10.1016/S1353-8020(11)70027-X.
3. Helmich RC, Hallett M, Deuschl G, Toni I, Bloem BR. Cerebral causes and consequences of Parkinsonian resting tremor: a tale of two circuits? *Brain*. 2012;135:3206–26. doi:10.1093/brain/aww023.
4. Clarimón J, Pagonabarraga J, Paisán-Ruiz C, Campolongo A, Pascual-Sedano B, Martí-Massó JF, et al. Tremor dominant Parkinsonism: clinical description and LRRK2 mutation screening. *Mov Disord*. 2008;23:518–23. doi:10.1002/mds.21771.
5. Jankovic J, McDermott M, Carter J, Gauthier S, Goetz C, Golbe L, et al. Variable expression of Parkinson's disease: a base-line analysis of the DATATOP cohort. The Parkinson Study Group. *Neurology*. 1990;40:1529–34.
6. Zetuský WJ, Jankovic J, Pirozzolo FJ. The heterogeneity of Parkinson's disease: clinical and prognostic implications. *Neurology*. 1985;35:522–6.
7. Williams-Gray CH, Foltynie T, Brayne CE, Robbins TW, Barker RA. Evolution of cognitive dysfunction in an incident Parkinson's disease cohort. *Brain*. 2007;130:1787–98. doi:10.1093/brain/awm111.
8. Selikhova M, Kempster PA, Revesz T, Holton JL, Lees AJ. Neuropathological findings in benign tremulous Parkinsonism. *Mov Disord*. 2013;28:145–52. doi:10.1002/mds.25220.
9. Prodoehl J, Planetta PJ, Kurani AS, Comella CL, Corcos DM, Vaillancourt DE. Differences in brain activation between tremor- and nontremor-dominant Parkinson diseases. *JAMA Neurol*. 2013;70:100–6. doi:10.1001/jamaneurol.2013.582.
10. Rosenberg-Katz K, Herman T, Jacob Y, Giladi N, Hendler T, Hausdorff JM. Gray matter atrophy distinguishes between Parkinson disease motor subtypes. *Neurology*. 2013;80:1476–84. doi:10.1212/WNL.0b013e31828cfaa4.
11. Zhang YH, Tang BS, Song CY, Xu Q, Lou MX, Liu ZH, et al. The relationship between the phenotype of Parkinson's disease and levodopa-induced dyskinesia. *Neurosci Lett*. 2013;556:109–12. doi:10.1016/j.neulet.2013.10.018.
12. Kang JH, Irwin DJ, Chen-Plotkin AS, Siderowf A, Caspell C, Coffey CS, et al. Association of cerebrospinal fluid β -amyloid 1–42, T-tau, P-tau181, and α -synuclein levels with clinical features of drug-naïve patients with early Parkinson disease. *JAMA Neurol*. 2013;70:1277–87. doi:10.1001/jamaneurol.2013.3861.
13. Spiegel J, Hellwig D, Samnick S, Jost W, Möllers MO, Fassbender K, et al. Striatal FP-CIT uptake differs in the subtypes of early Parkinson's disease. *J Neural Transm*. 2007;114:331–5. doi:10.1007/s00702-006-0518-2.
14. Helmich RC, Janssen MJ, Oyen WJ, Bloem BR, Toni I. Pallidal dysfunction drives a cerebellothalamic circuit into Parkinson tremor. *Ann Neurol*. 2011;69:269–81. doi:10.1002/ana.22361.
15. Schillaci O, Chiaravalloti A, Pierantozzi M, Di Pietro B, Koch G, Bruni C, et al. Different patterns of nigrostriatal degeneration in tremor type versus the akinetic-rigid and mixed types of Parkinson's disease at the early stages: molecular imaging with 123I-FP-CIT SPECT. *Int J Mol Med*. 2011;28:881–6. doi:10.3892/ijmm.2011.764.
16. Rossi C, Frosini D, Volterrani D, De Feo P, Unti E, Nicoletti V, et al. Differences in nigro-striatal impairment in clinical variants of early Parkinson's disease: evidence from a FP-CIT SPECT study. *Eur J Neurol*. 2010;17:626–30. doi:10.1111/j.1468-1331.2009.02898.x.
17. Isaias IU, Benti R, Cilia R, Canesi M, Marotta G, Gerundini P, et al. [123I]FP-CIT striatal binding in early Parkinson's disease patients with tremor vs. akinetic-rigid onset. *Neuroreport*. 2007;18:1499–502.
18. Song IU, Chung YA, Oh JK, Chung SW. An FP-CIT PET comparison of the difference in dopaminergic neuronal loss in subtypes of early Parkinson's disease. *Acta Radiol*. 2014;55:366–71. doi:10.1177/0284185113498075.
19. Eggers C, Kahraman D, Fink GR, Schmidt M, Timmermann L. Akinetic-rigid and tremor-dominant Parkinson's disease patients show different patterns of FP-CIT single photon emission computed tomography. *Mov Disord*. 2011;26:416–23. doi:10.1002/mds.23468.
20. Varrone A, Dickson JC, Tossici-Bolt L, Sera T, Asenbaum S, Booij J, et al. European multicentre database of healthy controls for [(123)I]FP-CIT SPECT (ENC-DAT): age-related effects, gender differences and evaluation of different methods of analysis. *Eur J Nucl Med Mol Imaging*. 2013;40:213–27. doi:10.1007/s00259-012-2276-8.
21. Pirker W. Correlation of dopamine transporter imaging with Parkinsonian motor handicap: how close is it? *Mov Disord*. 2003;18 Suppl 7:S43–51. doi:10.1002/mds.10579.
22. Fishman PS. Paradoxical aspects of Parkinsonian tremor. *Mov Disord*. 2008;23:168–73. doi:10.1002/mds.21736.
23. Mure H, Hirano S, Tang CC, Isaias IU, Antonini A, Ma Y, et al. Parkinson's disease tremor-related metabolic network: characterization, progression, and treatment effects. *Neuroimage*. 2011;54:1244–53. doi:10.1016/j.neuroimage.2010.09.028.
24. Hallett M. Tremor: pathophysiology. *Parkinsonism Relat Disord*. 2014;20 Suppl 1:S118–22. doi:10.1016/S1353-8020(13)70029-4.
25. Burn DJ, Rowan EN, Allan LM, Molloy S, O'Brien JT, McKeith IG. Motor subtype and cognitive decline in Parkinson's disease, Parkinson's disease with dementia, and dementia with Lewy bodies. *J Neurol Neurosurg Psychiatry*. 2006;77:585–9. doi:10.1136/jnnp.2005.081711.
26. Alves G, Larsen JP, Emre M, Wentzel-Larsen T, Aarsland D. Changes in motor subtype and risk for incident dementia in

- Parkinson's disease. *Mov Disord.* 2006;21:1123–30. doi:[10.1002/mds.20897](https://doi.org/10.1002/mds.20897).
27. Burn DJ, Landau S, Hindle JV, Samuel M, Wilson KC, Hurt CS, et al. Parkinson's disease motor subtypes and mood. *Mov Disord.* 2012;27:379–86. doi:[10.1002/mds.24041](https://doi.org/10.1002/mds.24041).
28. Kaasinen V, Rinne JO. Functional imaging studies of dopamine system and cognition in normal aging and Parkinson's disease. *Neurosci Biobehav Rev.* 2002;26:785–93.
29. Di Giuda D, Camardese G, Bentivoglio AR, Cocciolillo F, Guidubaldi A, Pucci L, et al. Dopaminergic dysfunction and psychiatric symptoms in movement disorders: a 123I-FP-CIT SPECT study. *Eur J Nucl Med Mol Imaging.* 2012;39:1937–48. doi:[10.1007/s00259-012-2232-7](https://doi.org/10.1007/s00259-012-2232-7).
30. Korchounov A, Schipper HI, Preobrazhenskaya IS, Kessler KR, Yakhno NN. Differences in age at onset and familial aggregation between clinical types of idiopathic Parkinson's disease. *Mov Disord.* 2004;19:1059–64. doi:[10.1002/mds.20061](https://doi.org/10.1002/mds.20061).
31. Marras C, Lang A. Parkinson's disease subtypes: lost in translation? *J Neurol Neurosurg Psychiatry.* 2013;84:409–15. doi:[10.1136/jnnp-2012-303455](https://doi.org/10.1136/jnnp-2012-303455).
32. Stebbins GT, Goetz CG, Burn DJ, Jankovic J, Khoo TK, Tilley BC. How to identify tremor dominant and postural instability/gait difficulty groups with the movement disorder society unified Parkinson's disease rating scale: comparison with the unified Parkinson's disease rating scale. *Mov Disord.* 2013;28:668–70. doi:[10.1002/mds.25383](https://doi.org/10.1002/mds.25383).
33. Liscic RM, Srulijes K, Gröger A, Maetzler W, Berg D. Differentiation of progressive supranuclear palsy: clinical, imaging and laboratory tools. *Acta Neurol Scand.* 2013;127:362–70. doi:[10.1111/ane.12067](https://doi.org/10.1111/ane.12067).
34. Oh M, Kim JS, Kim JY, Shin KH, Park SH, Kim HO, et al. Subregional patterns of preferential striatal dopamine transporter loss differ in Parkinson disease, progressive supranuclear palsy, and multiple-system atrophy. *J Nucl Med.* 2012;53:399–406. doi:[10.2967/jnumed.111.095224](https://doi.org/10.2967/jnumed.111.095224).
35. Savica R, Grossardt BR, Bower JH, Ahlskog JE, Rocca WA. Incidence and pathology of synucleinopathies and tauopathies related to Parkinsonism. *JAMA Neurol.* 2013;70:859–66. doi:[10.1001/jamaneurol.2013.114](https://doi.org/10.1001/jamaneurol.2013.114).
36. Kaundlstorfer C, Granata R, Wenning GK. Tremor in multiple system atrophy – a review. *Tremor Other Hyperkinet Mov (N Y).* 2013;3: <http://tremorjournal.org/article/view/165>.